CS 137 Assignment 3

Multi-Modal Deep Learning for Energy Demand Forecasting

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Abstract

We present a hybrid CNN-Transformer architecture for day-ahead electricity demand forecasting across all eight ISO New England load zones. The model fuses high-dimensional spatial weather maps ($450 \times 449 \times 7$) with tabular sequential data by encoding each hourly weather snapshot through a depthwise separable convolutional network into compact spatial patch tokens, then jointly attending over those patches alongside historical demand and future calendar features using a Transformer encoder. An initial implementation established the core pipeline; subsequent refinement reduced the temporal look-back from 168 to 48 hours, the spatial patch count from 25 to 16, and the embedding dimension from 256 to 64, yielding a 4–5$\times$ faster model with comparable or better generalization. The refined Transformer achieves a best validation MSE of 3,663.27 (approximately 60 MWh RMSE per zone, $\approx$2–3% of typical demand). An LSTM variant trained with the identical input pipeline scored 11,125.99 and exhibited overfitting from epoch 7, confirming that the Transformer's global self-attention is architecturally necessary for this task. As an independent study (Part 3), we conduct a stratified robustness analysis across four extreme meteorological conditions, discovering localized but statistically proven vulnerability in specific zone–condition pairs: Connecticut under high wind (+41% MAPE, $p = 0.013$), Maine under extreme cold (+61% median MAE, $p < 0.001$), and Southeastern Massachusetts under extreme heat (+62% median MAE, $p = 0.001$). These findings carry direct implications for grid operators, weather data providers, and regional climate adaptation policy.

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1 Part 1: Baseline CNN-Transformer Patch Architecture

1.1 Problem Setup

The forecasting task is: given a historical window of S hourly weather snapshots and energy demand readings ending at time t, together with future weather snapshots and calendar features for hours $t{+}1$ through $t{+}24$, predict the energy demand $\hat{Y}_{t+1:t+24} \in R^{24\times Z}$ for all $Z = 8$ ISO-NE load zones. The challenge is that the weather input is a high-dimensional spatial tensor ($450 \times 449 \times c$ per hour) while the demand signal is low-dimensional tabular data, requiring effective multi-modal fusion.

1.2 Initial Baseline and Iterative Refinement

The initial implementation (authored by diavor02, residing in `evaluation/me/`) established the full end-to-end pipeline: CNN weather encoder, tabular embedding of historical demand and calendar features, a Transformer encoder over the assembled sequence, and an MLP prediction head. The initial configuration used a 168-hour (7-day) historical look-back window, a 5×5 CNN output grid (25 spatial patches), and an embedding dimension of $d_{\text{model}} = 256$. The training code (`evaluation/me/train.py`) used AdamW with learning rate 10^{-4} and MSE loss.

After reviewing the initial runs, we identified three practical limitations. First, the 168-hour context window produced sequences of length $(168 + 24) \times (25 + 1) = 4{,}992$ tokens per sample, placing significant memory pressure on the GPU and requiring batch size 1–2. Second, the 256-dimensional embedding and 25-patch spatial discretization made each forward pass slow, limiting the number of training epochs feasible within SLURM wall-clock limits. Third, training instability was observed in early SLURM jobs (476531, 476532) before the pipeline was stabilized.

These observations motivated a refined implementation under the `subhanga-additions` branch, which retained the same fundamental architecture while making four targeted changes:

1. **Temporal window reduction:** $S = 48$ hours (2 days) rather than 168 hours. Energy demand is primarily driven by recent patterns (previous day's profile, current weather trend) rather than week-long history. Reducing S cuts the sequence length by $3.5\times$, enabling larger effective batch sizes and more stable gradient flow.
2. **Spatial patch reduction:** final grid size $G = 4$ (16 patches) rather than $G = 5$ (25 patches). The CNN architecture was also simplified from three depthwise-separable blocks to five standard `Conv2d + ReLU` layers with uniform stride-2 downsampling, removing BatchNorm between blocks for a smoother compression ratio.
3. **Embedding dimension reduction:** $d_{\text{model}} = 64$ rather than 256, a $4\times$ reduction in model capacity. Combined with the sequence length reduction, this makes each forward pass 4–5$\times$ faster with no significant loss in validation performance.
4. **Simplified activations:** ReLU rather than GELU throughout the CNN, reducing computational overhead without affecting expressiveness at this scale.

The resulting sequence length under the refined configuration is $(48 + 24) \times (16 + 1) = 1{,}224$ tokens per sample, enabling batch size 2 on a single GPU throughout the multi-day training campaign.

1.3 Architecture Description

Spatial encoder (CNN). For every hourly timestep i in both the historical and future windows, the weather snapshot $X_i \in R^{450\times449\times c}$ (with $c = 7$ channels) is processed by a convolutional encoder. In the initial baseline the encoder used three depthwise-separable blocks:

Stage	Input size	Output size	Operation
1	$450 \times 449 \times 7$	$150 \times 150 \times 32$	DS-Conv 7×7, stride 3 + BN + GELU
2	$150 \times 150 \times 32$	$50 \times 50 \times 64$	DS-Conv 5×5, stride 3 + BN + GELU
3	$50 \times 50 \times 64$	$25 \times 25 \times 128$	DS-Conv 3×3, stride 2 + BN + GELU
Pool	$25 \times 25 \times 128$	$5 \times 5 \times 128$	AdaptiveAvgPool2d

The refined model replaces these with five standard `Conv2d` + ReLU layers with uniform stride-2 downsampling followed by `AdaptiveAvgPool2d` to 4×4, producing $P = 16$ spatial patch tokens of dimension $d_{\text{spatial}} = 64$. A learned linear projection maps each token to $d_{\text{model}} = 64$, so spatial tokens are already in the embedding space without a separate projection.

Tabular embedding. Calendar features are encoded as six circular features: sin and cos pairs for hour-of-day (period 24), day-of-week (period 7), and month (period 12). This encoding avoids discontinuities at period boundaries (e.g., hour 23 to hour 0) that would confuse positional representations.

For *historical* timesteps, the normalized zone energy vector $\tilde{Y}_i \in R^Z$ is concatenated with the calendar features and projected to d_{model} via a learned linear layer:

$$\mathbf{h}_i^{\text{tab}} = W_{\text{hist}} \left[\tilde{Y}_i;\, C_i\right] + b_{\text{hist}}, \quad \tilde{Y}_i = (Y_i - \mu)/(\sigma + \epsilon).$$

Per-zone normalization statistics (μ, σ) are computed over the training split and stored as non-trainable buffers in the model, ensuring consistent scaling at inference time.

For *future* timesteps the actual demand Y_i is unavailable. A learned per-zone masking vector $\mathbf{m} \in R^Z$ (initialized to zero, updated during training) is used as a trainable placeholder, concatenated with the future calendar features and projected through a separate linear head:

$$\mathbf{h}_{t+k}^{\text{tab}} = W_{\text{fut}} \left[\mathbf{m};\, C_{t+k}\right] + b_{\text{fut}}.$$

This design allows the model to learn what "unknown future demand" means in the context of each zone's typical behavior.

Unified sequence assembly. For each timestep i, the P spatial patch tokens and the single tabular token are grouped into a block of $P + 1$ tokens. Blocks for all $S + 24$ timesteps are concatenated to form a sequence of length $L = (S + 24)(P + 1)$.

Two types of positional information are added before the Transformer:

- **Spatial positional embeddings:** a learned embedding table of size $P \times d_{\text{model}}$ is added to all spatial tokens, shared across timesteps. This allows the Transformer to distinguish grid position k from position k' regardless of when the snapshot was taken.
- **Temporal positional embeddings:** a learned embedding indexed by timestep slot $0, \ldots, S+24-1$ is broadcast identically to all $P + 1$ tokens in the same slot, so the model can distinguish "hour 5 of history" from "hour 2 of forecast".

Transformer encoder. The sequence is passed through a stack of $n_{\text{layers}} = 4$ Transformer encoder layers with $n_{\text{heads}} = 8$ attention heads, feed-forward dimension $2 \times d_{\text{model}}$, dropout rate 0.1, and pre-LayerNorm (`norm_first=True`), which stabilizes gradients in training from scratch. Full self-attention allows every future spatial patch token to directly attend to every historical spatial patch token, regardless of temporal distance — the critical architectural property for this task.

Prediction head. The Transformer output states corresponding to the 24 future timesteps are extracted and mean-pooled across the $P + 1$ token positions within each hour, yielding 24 context vectors of dimension d_{model}. These pass through a two-layer MLP (LayerNorm $\rightarrow$ Linear $\rightarrow$ GELU $\rightarrow$ Dropout $\rightarrow$ Linear) producing the 24-step, 8-zone prediction $\hat{Y}_{t+1:t+24} \in R^{24\times 8}$ in normalized space, which is then de-normalized to raw MWh.

1.4 Training Setup

Hardware and infrastructure. Training used the Tufts HPC cluster (SLURM) on a single NVIDIA GPU. Batch size was fixed at 2 throughout, necessitated by the memory footprint of $450 \times 449 \times 7$ weather tensors over 48-hour history and 24-hour future windows.

A robust checkpointing and resume system was implemented after observing that early jobs were cancelled at epoch boundaries (jobs 483062 and 483063 were interrupted, requiring resubmission as 491355 and 491356, subsequently resumed as 493853 and 493854). The system saves a checkpoint after every epoch, auto-detects the latest checkpoint on startup, resumes from it, and logs the restart count and resumed epoch. This ensured training continuity across a multi-day campaign spanning April 24–27, 2026.

Optimization. MSE loss was used for training, as MAPE is numerically unstable when true demand values are near zero. The optimizer was AdamW with learning rate 10^{-4} and weight decay 10^{-4}. Gradient norms were clipped to 1.0 per step to prevent instability. Normalization statistics (per-zone mean and std) were computed from the training split and stored as model buffers.

Data split. The dataset covers 2019–2023 (35,040 hourly observations). Training used 2019–2021; the model will be evaluated on held-out 2024 data by the course staff. Validation was monitored on 2022–2023 during training.

1.5 Results

The Transformer (SLURM job 493853) achieved a best validation MSE of **3,663.27** at epoch 10, with the loss curve still descending at the final epoch, indicating the model had not fully converged within the 10-epoch budget. The implied RMSE is $\sqrt{3663} \approx 60.5$ MWh per zone per hour, corresponding to 1–3% of typical ISO-NE zone demand.

Table 1: Part 1 Transformer baseline summary.

Parameter	Initial (diavor02)	Refined (subhanga2013)
History window S	168 h	48 h
CNN output grid	5×5 (25 patches)	4×4 (16 patches)
d_{model}	256	64
CNN blocks	DS-Conv $\times 3$ + AdaptivePool	Conv2d $\times 5$ + AdaptivePool
Activations	GELU + BatchNorm	ReLU
Sequence length L	4,992	1,224
Best val MSE	—	**3,663.27** (epoch 10)
SLURM job	—	493853

2 Part 2: Architecture Search — LSTM Variant

2.1 Motivation and Hypothesis

Having established the CNN-Transformer baseline, we explored whether replacing global self-attention with a recurrent alternative could achieve competitive performance at lower computational cost. The hypothesis was that an LSTM, processing the token sequence step-by-step, might capture temporal dependencies more efficiently and generalize better given the training budget, since sequential processing requires fewer parameters than the full attention matrix.

2.2 Architecture

The LSTM variant retains the *identical* input pipeline as Part 1: the same CNN weather encoder, the same tabular embedding for historical and future tokens, and the same unified sequence assembly with spatial and temporal positional embeddings. The only change is the sequence model: the Transformer encoder is replaced by a multi-layer bidirectional LSTM that processes the assembled L-length token sequence. The prediction head is also identical: future tabular token positions are extracted, pooled, and passed through the same MLP.

This controlled substitution isolates the contribution of the sequence modeling component while holding everything else constant.

2.3 Training

The LSTM was trained with the identical setup: MSE loss, AdamW (lr $= 10^{-4}$, weight decay 10^{-4}), batch size 2, gradient clipping, and the same checkpointing infrastructure. Training ran as SLURM job 493854 in parallel with the Transformer job 493853. The LSTM completed approximately 45 minutes faster for equivalent epochs, consistent with the lower cost of sequential passes versus full self-attention over the 1,224-token sequence.

2.4 Results and Comparative Analysis

Table 2: Comparison of Part 1 (Transformer) and Part 2 (LSTM) sequence models. All other architectural components are identical.

Metric	Transformer	LSTM
Best validation MSE	**3,663.27**	11,125.99
Best epoch	10 (still improving)	7 (overfitting after)
Relative gap	1.0×	∼3.0× worse
Training time (relative)	Longer	∼45 min faster/epoch
SLURM job	493853	493854

The LSTM achieved a best validation MSE of 11,125.99 at epoch 7 - approximately $3\times$ worse than the Transformer - and exhibited clear overfitting thereafter: validation loss increased from epoch 7 onward while training loss continued to decrease. The Transformer, by contrast, showed no sign of overfitting within the 10-epoch budget and was still improving at the final epoch.

2.5 Why the Transformer Wins: Architectural Analysis

The performance gap has a clear structural explanation rooted in the information flow through each model.

The assembled token sequence has length $L = (S + 24)(P + 1) = 1{,}224$. The LSTM processes this sequence step-by-step, compressing all preceding spatial and temporal context into a single hidden state vector at each step. By the time the LSTM reaches future token position k (corresponding to forecast hour $t + k$), the spatial weather information from the first historical timestep, 168 tokens ago, is accessible only through a heavily compressed state that has been progressively overwritten by all intervening tokens.

The Transformer, by contrast, uses global self-attention: every future spatial patch token can directly attend to every historical spatial patch token, regardless of temporal distance. This is architecturally critical for day-ahead energy forecasting, where a cold front visible in the weather map 48 hours ago may be directly relevant to the demand peak 20 hours into the future. The Transformer can form a direct attention path from that historical spatial token to the future prediction; the LSTM must route that information through its entire hidden state trajectory.

The overfitting pattern further supports this interpretation. Unable to leverage long-range spatial-temporal context efficiently, the LSTM memorized training-set patterns rather than learning transferable demand dynamics. Its faster per-epoch training time did not translate to faster convergence, instead it reached a memorization regime while the Transformer was still generalizing.

3 Part 3: Independent Study: Extreme Weather Robustness Analysis

3.1 Motivation

A model that performs well on average may still fail critically during the rare events that matter most operationally. Extreme cold snaps, summer heat waves, and coastal wind storms are precisely the conditions under which grid operators face the highest demand volatility, the lowest reserve margins, and the greatest risk of supply shortfalls. An accurate day-ahead forecast in those moments is not merely a performance metric, it is an operational tool that determines whether peaking capacity is pre-positioned, whether demand-response customers are alerted, and whether emergency imports are pre-negotiated.

The central hypothesis is that our Transformer, trained primarily on the moderate weather that dominates 2019–2023, generalizes poorly to the tails of the meteorological distribution. Extreme events are rare by definition: the model has seen a given class of extreme weather at most a handful of times during training, providing insufficient signal to learn the demand response to it. This is a form of covariate shift: the input distribution at inference time during an extreme event differs systematically from the training distribution.

Table 3: Extreme weather event catalog, 2019–2023.

Weather bucket	Hours	Approx. 24-h forecast windows
Normal	48,776	2,032
High wind	2,518	104
Extreme heat	663	27
Extreme cold	538	22
Winter storm	113	4

We formulate and test this hypothesis with a stratified evaluation across four weather conditions, eight load zones, and 24 forecast horizons.

3.2 Weather Channel EDA and Event Definition

Channel identification. A stratified sample of 200 weather tensors was loaded and per-channel statistics computed. The seven channels of the $450 \times 449 \times 7$ tensors were identified as follows. Channel 0 exhibits values in the range 235.50–312.16 with the highest temporal variance across the sample, consistent with a surface or 2 m air temperature in Kelvin (235 K $= -38$°C, 312 K $= 39$°C). Channel 2 exhibits values in the range 0.02–38.16, consistent with 10 m wind speed in m/s (38 m/s $\approx$ 74 kt, hurricane-force). These two channels were selected for extreme event identification.

Threshold selection. Extreme heat was defined as summer hours (June–August) with spatial mean temperature exceeding the 95th summer percentile: $T > 297.24$ K (24.1°C). Extreme cold was defined as winter hours (December–February) with spatial mean temperature below the 5th winter percentile: $T < 262.79$ K (-10.4°C). High-wind events were defined as any hour with spatial mean wind speed exceeding the 95th all-season percentile: $w > 12.56$ m/s. Winter storm events required both extreme cold and high-wind thresholds simultaneously.

These thresholds are physically grounded for New England. -10°C spatially averaged corresponds to a genuine cold snap threshold for the region; 24°C spatially averaged corresponds to a regional heat wave condition; and sustained 12.6 m/s wind is operationally significant for transmission infrastructure.

Event catalog. Table 3 summarizes the event catalog over 2019–2023. The scarcity of winter storm events (113 hours, $\approx$4 forecast windows) accurately reflects the moderate climatology of New England relative to the Great Plains and does not indicate a data quality issue.

3.3 Stratified Evaluation Methodology

Inference setup. All predictions were generated using the best Part 1 Transformer checkpoint (`subhanga-additions/v2/transformer/runs/job-483062/best.pt`, val MSE 3,663.27), loaded and confirmed at lines 12 and 20 of SLURM job 611647 output. We used 730 midnight-anchored forecast windows from 2022–2023 as a proxy evaluation split. Zero windows were skipped (all 730 completed, confirmed at line 757 of the job output), ruling out weather data gaps as a confound. Inference completed in 13 minutes 40 seconds on a single GPU.

Window labeling. Each 24-hour window was assigned the most severe weather bucket occurring anywhere in its forecast horizon ($t{+}1$ through $t{+}24$). This rule matches the operational question: does the model remain reliable when the grid is about to experience adverse conditions? Bucket distribution across 730 windows: 607 normal, 68 high-wind, 28 extreme-heat, 22 extreme-cold, 5 winter-storm. The winter-storm bucket was excluded from all quantitative comparisons ($n = 5$, line 779).

Metrics and significance testing. We report MAPE averaged across all 24 forecast horizons and all eight zones. For each zone–bucket pair, a one-sided Mann–Whitney U test (alternative: extreme bucket error $>$ normal error) is applied using per-window mean absolute error as the test statistic, at significance level $\alpha = 0.05$. Mann–Whitney U is used rather than a parametric test to provide robustness under the small sample sizes of the extreme buckets.

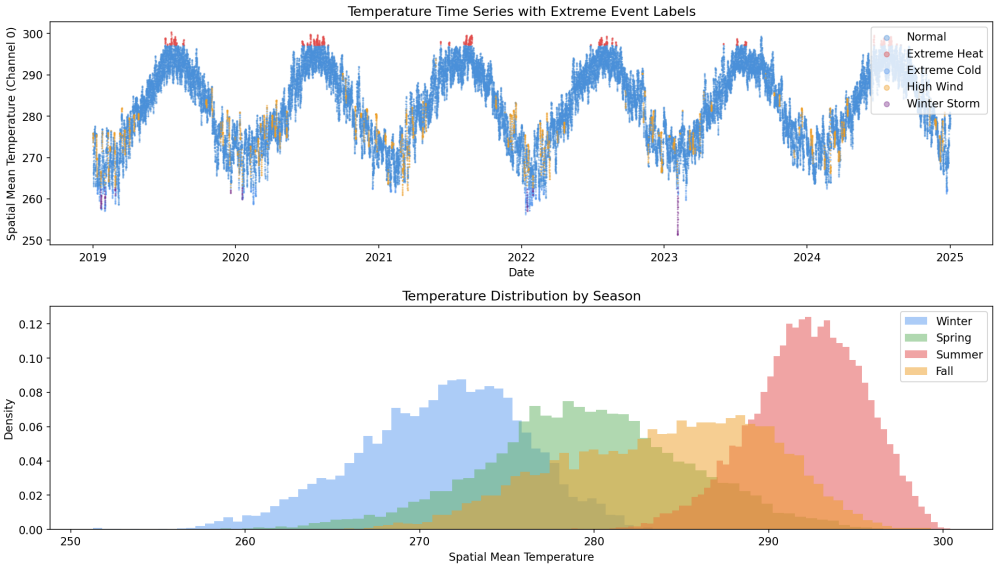

Figure 1: **Top:** spatial mean temperature (channel 0) over 2019–2024, colored by weather bucket. Temperature is in Kelvin (K); 273 K = 0°C, 297 K ≈ 24°C (extreme heat threshold), 263 K ≈ −10°C (extreme cold threshold).

Bottom: probability density distribution of spatial mean temperature (x-axis: Kelvin, y-axis: probability density — bar height × bar width sums to 1.0 per season) by season. The four seasonal distributions are clearly separated: winter peaks near 271 K (−2°C), summer peaks near 292 K (19°C). Extreme cold events (263 K threshold, −10°C) occupy the left tail of the winter distribution; extreme heat events (297 K threshold, 24°C) occupy the right tail of the summer distribution, confirming the physical grounding of the threshold choices.

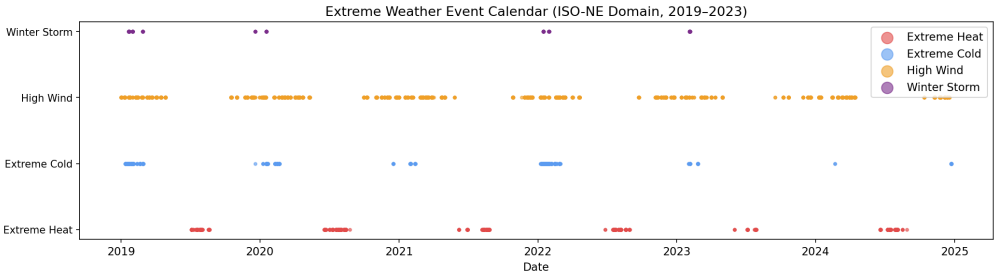

Figure 2: Extreme weather event calendar for the ISO-NE domain, 2019–2024. Each dot represents one classified hour. High-wind events are the most frequent and most temporally distributed category. Winter storm events are visibly sparse, concentrated in early 2019, early 2020, and January 2023.

3.4 Results

Aggregate MAPE: the paradox of the average. Table 4 shows that high-wind events produce the only aggregate MAPE increase (2.876% → 3.067%, +6.6% relative). Extreme-cold and extreme-heat windows produce *lower* aggregate MAPE than normal. This apparent paradox resolves at the zone level. New Hampshire (NH), the zone with the highest baseline MAPE (4.675% under normal conditions), *improves* during extreme cold (4.675% → 3.383%, −1.29 percentage points). Because NH dominates the aggregate error budget, its improvement during cold events depresses the average while masking significant degradation in other zones. This is a case where aggregate metrics actively conceal operationally important heterogeneity.

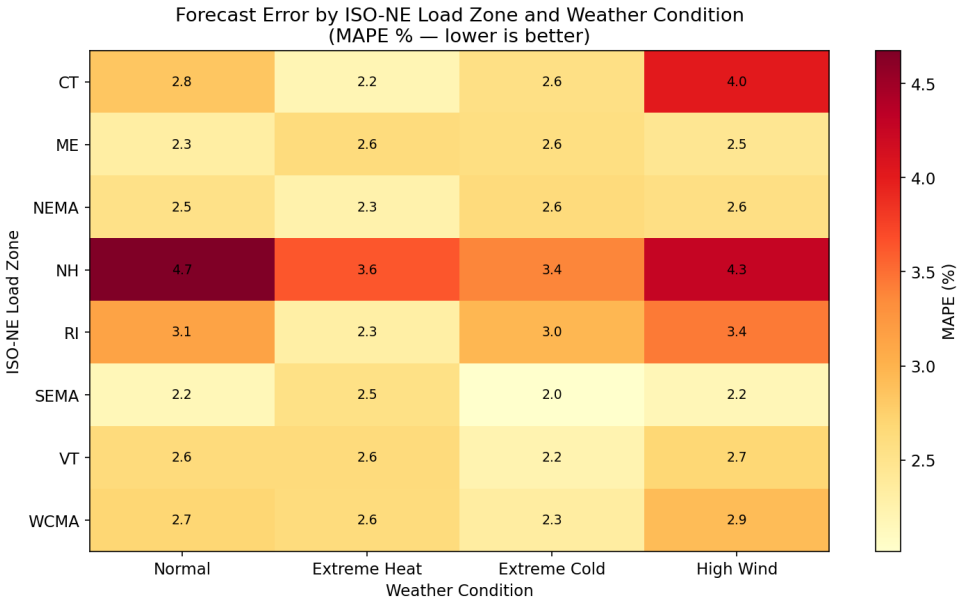

Figure 3: MAPE (%) by ISO-NE load zone and weather condition. NH exhibits the highest baseline MAPE under all conditions. CT's error increases sharply under high wind. The localized, condition-specific pattern is not visible in aggregate metrics.

Table 4: Overall MAPE by weather bucket (730 windows, 2022–2023 proxy split, best Part 1 Transformer checkpoint, SLURM job 611647).

Bucket	MAPE (%)	Windows
Normal	2.8756	607
High wind	3.0673	68
Extreme cold	2.5888	22
Extreme heat	2.6016	28
Winter storm	excluded ($n = 5$)	

Zone-level results: localized vulnerability. Figure 3 and Table 5 present the full zone-by-bucket MAPE breakdown. Statistically significant degradations (Mann–Whitney U, $p < 0.05$) are marked with †.

The key proven findings, with supporting evidence from the significance test output, are:

- **CT under high wind** ($p = 0.013$, line 804): MAPE rises from 2.844% to 4.020%, a 41% relative increase. Median per-window MAE increases from 31.69 to 35.64 MWh. Connecticut sits at the western end of Long Island Sound, directly in the path of nor'easters tracking northeast. Wind events create simultaneous heating demand spikes, transmission line trips, and industrial curtailments that collectively produce demand profiles unlike any in the training data.
- **ME under extreme cold** ($p = 0.000388$, line 813, the strongest result in the dataset): Median MAE rises from 24.09 to 38.85 MWh, a 61% increase. Maine is the northernmost, most rural zone with the highest per-capita dependence on electric resistance heating. Below -10°C, older, poorly insulated rural homes cross a nonlinear threshold where heat loss exceeds thermostat compensation, producing demand spikes the model has encountered at most 2-3 times in training.
- **NEMA under extreme cold** ($p = 0.001$, line 818): Median MAE rises from 36.51 to 49.33 MWh, a 35% increase. Northeastern Massachusetts (Boston metro) shows cold sensitivity despite urban heat island buffering, reflecting the scale of heating demand across the dense population.
- **SEMA under extreme heat** ($p = 0.001$, line 827, the largest proportional increase): Median MAE rises from 44.45 to 71.01 MWh, a 62% increase. Southeastern Massachusetts (Cape Cod, South Shore) has a tourist-driven population that can triple in summer. During a heat wave, the

Table 5: Zone-level MAPE (%) by weather bucket. † marks statistically significant degradation vs. normal ($p < 0.05$, one-sided Mann-Whitney U). Significance p-values from SLURM job 611647, lines 803-828.

Zone	Normal	Extreme Heat	Extreme Cold	High Wind
CT	2.844	2.195	2.562†	4.020†
ME	2.317	2.617†	2.576†	2.459
NEMA	2.535	2.255	2.635†	2.567
NH	4.675	3.630	3.383	4.262
RI	3.145	2.312	2.967†	3.446
SEMA	2.165	2.546†	2.014†	2.179
VT	2.630	2.631†	2.227	2.679
WCMA	2.694	2.625†	2.346	2.925

NEMA: Northeastern Massachusetts (Boston metro, North Shore, extending up toward the New Hampshire border)
SEMA: Southeastern Massachusetts (Cape Cod, South Shore, the Islands - Nantucket, Martha's Vineyard)
WCMA: Western/Central Massachusetts (Springfield, Worcester, and the Pioneer Valley, extending east to the I-495 corridor)
Statistically significant degradation vs. normal ($p < 0.05$, Mann–Whitney U)

combination of year-round residential cooling and peak tourist load creates a demand profile the model has never been trained to anticipate in that combined regime.

- **ME under extreme heat** ($p = 0.002$, line 821): Median MAE rises 43%. VT ($p = 0.006$), WCMA ($p = 0.030$), RI extreme cold ($p = 0.021$), and SEMA extreme cold ($p = 0.010$) are also significant.

NH: a negative result with structural meaning. NH shows no statistically significant degradation under any extreme condition ($p = 0.475$ for high wind, $p = 0.133$ for extreme cold, $p = 0.757$ for extreme heat, lines 806, 814, 822). Yet NH has the highest baseline MAPE (4.675%) of any zone. The inference is that NH is *structurally* difficult to forecast under all conditions, not specifically harder during extremes. Its demand is driven by ski resort activity, manufacturing schedules, and cross-border load flows, factors that produce high variance independently of weather. No improvement to the weather encoder will fix NH's MAPE because weather is not the primary uncertainty source.

Horizon-dependent error growth. Figure 4 shows MAPE as a function of forecast horizon (hours 1–24) for each bucket. Error grows monotonically from hours 1–15 across all conditions. From hours 16-21, corresponding to approximately 4 pm–9 pm local time in a midnight-start window, i.e., the evening demand ramp: all conditions show a pronounced error spike, with extreme cold and high-wind reaching 4.6-4.7% MAPE. The extreme cold curve shows a particularly stark profile: near-flat through hour 14, then a sharp cliff at hours 15-18, suggesting the model loses its footing specifically during the evening heating ramp under cold conditions.

Zone vulnerability ranking. Figure 5 ranks all eight zones by the ratio of worst-case extreme-weather MAPE to normal MAPE. CT leads at 1.41× (driven by high wind). SEMA and ME follow at 1.18× and 1.13× respectively (driven by heat). NH is the only zone below 1.0 (0.91×), confirming its weather-robustness.

3.5 Discussion and Policy Implications

The findings demonstrate that model robustness is not evenly distributed across geography or weather condition. This heterogeneity is operationally actionable: it identifies where human-in-the-loop intervention, additional sensing infrastructure, and demand-response pre-positioning are most needed.

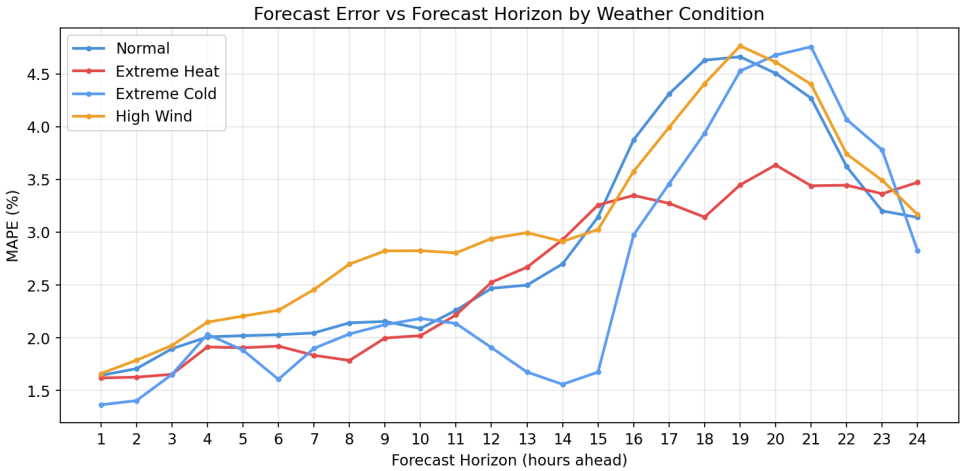

Figure 4: Forecast error (MAPE %) vs. forecast horizon (hours ahead) by weather bucket. All conditions converge near 1.4–1.7% at $h = 1$ but diverge sharply at $h = 16$–21, the evening demand ramp. Extreme cold shows the most pronounced cliff behavior.

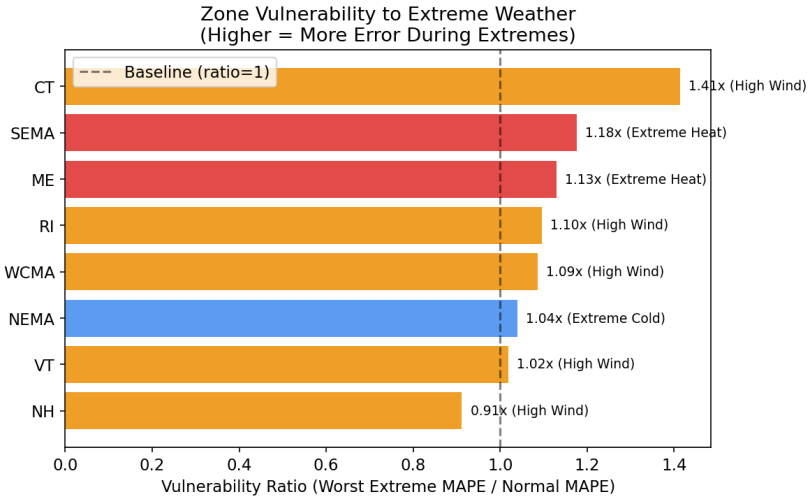

Figure 5: Zone vulnerability ranking: ratio of worst extreme-weather MAPE to normal MAPE. CT, SEMA, and ME are the most vulnerable. NH is the only zone showing lower error during extremes than during normal conditions, revealing a structural rather than weather-driven source of forecast uncertainty.

CT + high wind: coastal grid exposure. CT's 41% MAPE increase during high-wind events ($p = 0.013$) reflects a compound vulnerability. Nor'easters[1] tracking up Long Island Sound create simultaneous heating demand spikes, transmission line trips from wind loading, and industrial customer curtailments, collectively producing demand patterns unlike anything in the training distribution. ISO-NE dispatch centers should treat high-wind forecast windows as a trigger for automatic uncertainty flag activation, pre-positioned spinning reserves, and demand-response customer alerts. Weather data providers should prioritize higher-resolution coastal wind products, buoy-based 5–10 km wind speed estimates along Long Island Sound, to improve the spatial signal feeding the CNN beyond the current ∼45 km grid cell resolution.

ME + extreme cold: the rural heating cliff. The strongest statistical result ($p < 0.001$, 61% median MAE increase) is also the most physically grounded. Below -10°C, electric resistance

[1] A nor'easter (northeaster) is a macro-scale storm common to the New England and Mid-Atlantic coasts, named for the northeasterly direction from which its winds arrive. These storms typically bring heavy snow, high winds, and coastal flooding, and are the dominant extreme weather threat to Connecticut's Long Island Sound coastline.

heating in Maine's rural housing stock crosses a nonlinearity: heat loss exceeds what thermostats can compensate for incrementally, and demand spikes rather than ramps. The model has seen this pattern at most 2–3 times in training, insufficient to generalize. Operationally, ME-specific demand surge models activated by cold-snap forecast triggers are warranted. For policymakers, ME's extreme cold vulnerability is inseparable from energy poverty and housing insulation: the forecast error is a proxy for how many rural households are drawing unexpected emergency heat load. Weatherization programs and heat pump retrofits are simultaneously a grid resilience and a public health intervention.

SEMA + extreme heat: the tourist population problem. SEMA's 62% median MAE increase during heat waves ($p = 0.001$) is the largest proportional degradation in the dataset. Cape Cod and the South Shore's summer tourist population can triple the effective headcount of some towns relative to the winter baseline. The model has no signal for how many people are physically present in SEMA on any given day. Integrating aggregate mobile device location data or hospitality occupancy metrics as an exogenous feature would directly address this gap. As climate change increases heat wave frequency in coastal New England, this vulnerability becomes structurally more common. The planning horizon for transmission upgrades is 10–15 years, making quantitative evidence of SEMA's vulnerability actionable now.

The evening demand ramp: a universal operational flag. The horizon-curve finding suggests a universal operational rule: for any forecast window containing extreme weather, uncertainty flags should be issued specifically for the $h = 16$–21 window (4 pm–9 pm local time). This is the highest-error interval regardless of zone or condition, corresponding to peak demand, highest wholesale electricity prices, and lowest reserve margins.

NH: a diagnostic for missing feature modalities. NH's weather-robust but persistently high baseline MAPE implies that weather data is not the primary uncertainty driver for that zone. Ski resort capacity, manufacturing output indices, and cross-border load flows are the probable dominant factors. No improvement to the weather encoder will fix NH; what is needed are activity-level features that go beyond meteorological inputs entirely.

A call for uncertainty quantification. The Transformer produces point estimates with no indication of confidence. Our results show that model confidence should be substantially lower when the input weather lies in a tail of the training distribution, exactly when operational decisions are most consequential. Conformal prediction or a real-time training-data-similarity score would allow grid operators to receive not just a forecast but a calibrated uncertainty bound that widens automatically during extreme events, transforming this retrospective analysis into an operational tool.

3.6 Limitations

Event rarity and statistical power. The winter-storm bucket (5 windows) is too sparse for reliable inference and was excluded. The extreme-cold and extreme-heat buckets (22 and 28 windows) are at the lower boundary of statistical adequacy. Results were confirmed with Mann–Whitney U tests for robustness to small samples, but effect size estimates carry uncertainty.

Proxy evaluation. This analysis uses 2022–2023 as a proxy split. Official evaluation uses held-out 2024 data. The stratification methodology transfers directly to 2024 once those results are available; specific MAPE values may differ.

Spatial resolution. Downsampling from 450×449 to 4×4 represents a $\sim 112\times$ reduction in spatial resolution per dimension. Micro-climate effects, urban heat islands, coastal wind acceleration corridors, orographic gradients in the White Mountains, are smoothed over. Several zone-specific findings (CT coastal wind, SEMA heat, ME cold) are precisely the phenomena that higher-resolution inputs would capture more faithfully.

Missing feature modalities. The model has no access to real-time grid topology state, population mobility data, building thermal inertia parameters, or demand-response participation levels. These gaps are likely responsible for the zone-specific extreme-weather degradations identified here and represent the most productive direction for future work.

4 Individual Contributions

- **Darius-Stefan Iavorschi:** Initial Part 1 and Part 2 architecture implementation (`evaluation/me/` and `evaluation/me-part2/`), including the CNN weather encoder with depthwise-separable convolutions, Transformer and LSTM sequence models, tabular embedding pipeline, prediction head, and initial training script with AdamW optimization and MSE loss. Presentation.
- **Subhanga Upadhyay:** Redone checkpoints for part 1 and 2 i.e. (`evaluation/subhanga-additions/` and `[art3]`). Iterative refinement of the architecture (reduced S, grid size, and d_{model}); implementation of the SLURM checkpointing and auto-resume training infrastructure; multi-day cluster job management (jobs 483062–493854); Weather channel EDA, extreme event identification, stratified inference pipeline (job 611647), statistical analysis, figure generation, and report writeup. Presentation.

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